USN | | | | | | | | | | [22CSE154]

# THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

## COMPUTER ARCHITECTURE

**Time: 3 Hours** Maximum**Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

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| | | **PART-A** | **5 X 2=10** | | |
| **1.** | **a.** | Write the difference between Multiplexer and De- multiplexer | L1 | CO1 | **02** |
| | **b.** | What is selective set micro operation? | L1 | CO3 | **02** |
| | **c.** | Write the structure of instruction format. | L1 | CO4 | **02** |
| | **d.** | Point out the difference between I/O Mapped and Memory mapped I/O | L1 | CO5 | **02** |
| | **e.** | What is a volatile memory? Give an example. | L1 | CO6 | **02** |
| | | **PART-B** | **6 X 5=30** | | |
| **2.** | **a.** | Define Bus? Discuss the single bus structure with a neat diagram. | L2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | Explain the basic functional units of a computer with a neat block diagram. | L2 | CO1 | **05** |
| **3.** | | | | | |
| | **a.** | Differentiate between RISC and CISC Architecture. | L2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | In a computer the clock speed of the processor is 4GHz. Suppose you are executing a program with 100 instructions and each instruction takes 1.2 clock cycles to complete. What will be the execution time for the computer (in ns)? | L3 | CO1 | **05** |
| | | | | | |
| **4.** | **a.** | With an example explain the following logical shift micro operation. a. Logical Arithmetic shift b. Circular Shift | L2 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Illustrate single parity error checking code with an example. | L3 | CO3 | **05** |
| | | | | | |
| **5.** | **a.** | Interpret the phases of instruction cycle. | L3 | CO4 | **05** |
| | | **OR** | | | |
| | **b.** | Draw the hardware implementation block diagram of signed multiplication operation and explain it | L2 | CO4 | **05** |
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| **6.** | **a.** | Provide the details classification of parallel data transfer mode and compare parallel , serial data transfer mode | L2 | CO5 | **05** |
| | | **OR** | | | |
| | **b.** | Explain source initiated strobe control in Asynchronous data transfer methods | L2 | CO5 | **05** |
| | | | | | |
| **7.** | **a.** | Examine Characteristics of Memory Hierarchy Design | L3 | CO6 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss cache write policy | L2 | CO6 | **05** |
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| | | PART-C | 6 X 10=60 | | |
|---|---|---|---|---|---|
| 8. | a. | Illustrate the Properties of Excess-3 code and Convert thefollowing Decimal Numbers into Excess- 3 Code and to BCD as well. a. 40 b. 88 | L3 | CO2 | 10 |
| | | **OR** | | | |
| | b. | Simplify the following Boolean expression using K-Map:<br>F (A, B, C, D) = Σ (0,2,5,7,8,10,13,15)<br>F (A, B, C) = π (0,3,6,7) | L3 | CO1 | 10 |
| | | | | | |
| 9. | a. | Demonstrate the method of converting Binary Number into Gray code number with a suitable logic diagram. | L3 | CO2 | 10 |
| | | **OR** | | | |
| | b. | Design a composite logic circuit to perform basic logic operation AND,OR,NOT and EX-OR | L3 | CO3 | 10 |
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| 10. | a. | A digital computer has a common bus system for 8 registers of 8 bits each. The bus is constructed with multiplexers.<br>**i.** How many selection inputs are there in each multiplexer?<br>**ii.** What is the size of each multiplexer?<br>**iii.** How many multiplexers are there in the bus?<br>**iv.** Write the hardware implementation of the above logic circuit? | L3 | CO3 | 10 |
| | | **OR** | | | |
| | b. | Demonstrate with an example how the following micro operation performed<br>**i.** Selective-complement<br>**ii.** Selective-clear<br>**iii.** Mask (Delete)<br>**iv.** Clear<br>**v.** Insert | L3 | CO3 | 10 |
| | | | | | |
| 11. | a. | Demonstrate the direct and indirect addressing modes with a neat diagram. | L3 | CO4 | 10 |
| | | **OR** | | | |
| | b. | With help of booth algorithm explain multiplication two signed number (-3 and -4). | L2 | CO4 | 10 |
| | | | | | |
| 12. | a. | Name the three modes of data transfer and explain in detail any one mode. | L2 | CO5 | 10 |
| | | **OR** | | | |
| | b. | With neat a sketch explain the pipelining process. | L2 | CO5 | 10 |
| | | | | | |
| 13. | a. | Discuss Associative cache mapping. | L2 | CO6 | 10 |
| | | **OR** | | | |
| | b. | List the four cache replacement algorithm and explain any two with examples. | L2 | CO6 | 10 |